PAPER ID-310638

Subject Code: KAU072

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM VII) THEORY EXAMINATION 2024-25

HYBRID VEHICLE PROPULSION

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	Define the term Electric Vehicle.	1	K2
b.	Explain the term gradient resistance	1	K2
c.	Describe the concept of hybrid electric drivetrain	2	K2
d.	List the advantages and disadvantages of Electric Traction system	2	K1
e.	Explain the different types of wound field DC motor	3	K2
f.	Discuss the requirements of DC motors used in Electric Vehicle.	3	K2
g.	Discuss the basic structure of a fuel cell.	4	K2
h.	Describe the desirable properties of ultra-capacitors	4	K2
i.	Discuss the function of energy management systems in hybrid vehicle	5	K2
j.	Discuss in brief basic principle of deterministic control strategies.	5	K2

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a.	Explain impact of modern drive trains on energy supplies.	1	K2
b.	Describe the parallel hybrid topology along with its application	2	K2
c.	Describe the construction of BLDC motor with the help of neat sketch	3	K2
d.	Describe the working of Nickel-Metal Hydride battery along with its advantages and disadvantages.	4	K2
e.	Explain the challenges associated with the real-time implementation of energy management strategies in hybrid and electric vehicles	5	K2

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe in detail the history of hybrid and electric vehicles.	1	K2
b.	Explain the transmission characteristics of a vehicle.	1	K2

4. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe the power flow control in parallel hybrid with the help of neat sketches.	2	K2
b.	Discuss different considerations on which design of power control strategies depends.	2	K2

5. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe the Multi-quadrant control of DC motors used in hybrid electric vehicles	3	K2
b.	Describe the direct rotor flux orientation schemes of induction motor drives used in hybrid electric vehicles	3	K2

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6. Attempt any *one* part of the following: 10 x 1 = 10

a.	How does the integration of flywheel-based energy storage contribute to the overall performance of hybrid and electric vehicles.	4	K2
b.	In what ways does the safety of energy storage systems, particularly batteries, impact the design and operation of hybrid and electric vehicles	4	K2

7. Attempt any *one* part of the following: 10 x 1 = 10

a.	How do rule-based and optimization-based energy management strategies differ in their approach and application?	5	K2
b.	Describe the impact of energy management strategies on overall efficiency and performance of hybrid and electric vehicles.	5	K2

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